

$$\mathbf{e}_{t,k}^A = P_A \mathbf{W}_t^A \left(\mathbf{Y}_t^A - \hat{\mathbf{A}}_{t,k}^1 \right),$$

$$v_{\text{FBFM}}^A = v_{\theta_A}^A + \lambda_{\tau_k^A}^A \left(\mathbf{J}_{t,k}^A \right)^\top \mathbf{e}_{t,k}^A$$